

Rishav Gupta

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RESEARCH INTERESTS

Mobile health sensing and mobile systems, with an emphasis on acoustic and ultrasonic sensing, digital signal processing, and machine learning for cuff-free cardiovascular monitoring on commodity smartphones.

EDUCATION

- University of Maryland, Baltimore County** Jan. 2026 – Present
Ph.D. in Computer Science
• Advisor: [Dr. Dong Li](#), [Future Sensing and Interaction \(FSI\) Lab](#).
Baltimore, MD
- University of Maryland, Baltimore County** Jan. 2025 – Dec. 2025
M.S. in Computer Science
• GPA: 4.00/4.00. Coursework: Design and Analysis of Algorithms, Machine Learning, Neural Networks.
Baltimore, MD
- Manipal Institute of Technology** Jun. 2015 – May 2019
B.Tech. in Information Technology
• Ranked 8th out of 132 students in the graduating class.
Manipal, India

RESEARCH EXPERIENCE

- University of Maryland, Baltimore County** Jun. 2025 – Present
Graduate Research Assistant, Future Sensing and Interaction (FSI) Lab
Baltimore, MD
- **Smartphone blood-flow sensing** [2, 4]. Recovering arterial blood-flow waveforms and Doppler indices through the skin via phase-based ultrasonic Doppler; first author of the real-time demonstration [2].
 - **Acoustic tissue characterization** [1]. First-author work estimating soft-tissue relaxation time from smartphone acoustic chirps and echoes, validated against laboratory indentation (phantom and human forearm).
 - **Voice-scam protection** [3]. Contributed to Ultrasound Watermark, which binds live call audio to the caller's articulatory motion sensed with smartphone ultrasound.

PUBLICATIONS

- [1] **Rishav Gupta**, Alyssa N. Maguina, Sithumina Weerathna, Janelle P. Clark, and Dong Li. "WiP: Smartphone-based Acoustic Tissue Characterization for Personalized Body-contact Systems." In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Work-in-Progress Track. ACM, 2026. To appear.
- [2] **Rishav Gupta**, Shirui Cao, Jie Xiong, Sunghoon Ivan Lee, Jeremy Gummesson, and Dong Li. "Demo: Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones." In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Demo Track. ACM, 2026. To appear.
- [3] Renzhi Hao, Shirui Cao, Yizhu Wen, **Rishav Gupta**, Dharani Nadendla, Mehran Kafai, Hanqing Guo, and Dong Li. "Ultrasound Watermark: Real-time Acoustic Watermarking against Voice Scams on Smartphones." In *Proceedings of the 9th ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP '26)*, co-located with ACM/IEEE SEC. IEEE, 2026. To appear.
- [4] Shirui Cao, Jie Xiong, **Rishav Gupta**, Sunghoon Ivan Lee, Jeremy Gummesson, and Dong Li. "Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones." In *Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services (MobiSys '26)*, Emerging Ideas Track. ACM, 2026. [\[Paper\]](#) [\[Code\]](#)

HONORS AND AWARDS

COEIT Computing for Social Good Award , Research Day, UMBC • One of only two awards presented across 114 posters; conferred by Dean Jeanne M. van Briesen.	May 2026
First Place , Maryland NEW VENTURE™ Program Pitch Night, bwtech@UMBC • Awarded for <i>PocketCardio</i> , a smartphone-based cardiovascular screening venture.	May 2026
Best Poster Award , CSEE Research Day, UMBC	Apr. 2026
Best Poster Award , COEIT Research Day, UMBC	May 2025

INDUSTRY EXPERIENCE

Gojek (GoTo Group) <i>Senior iOS Engineer</i> • Reduced annual platform cost by \$3M by replacing a third-party analytics SDK (CleverTap) with an in-house, open-source tool at full feature parity. • Cut development time by 20% across the engineering team by automating the bi-weekly CI/CD release pipeline in Python and Shell on GitLab. • Doubled QA throughput with a real-time Event Visualizer for verifying analytics events in situ, and cut roughly 3 MB from the app binary through SwiftSyntax AST-based dead-code detection.	Feb. 2022 – Sep. 2024 Bengaluru, India
Nuclei <i>Senior iOS Developer and Team Lead</i> • Founding iOS developer of FedMobile (Federal Bank); implemented 60% of shipped features, including Payments, Investments, Loans, Onboarding, and Card Management. • Promoted to Team Lead; led four iOS developers and closed a four-month delivery gap between the iOS and Android platforms.	Jul. 2019 – Feb. 2022 Bengaluru, India
Hera doo <i>Android Developer Intern</i> • Built an end-to-end Android sales-order application from scratch, increasing field sales by 25%.	Jun. – Jul. 2018 Mostar, Bosnia

SERVICE AND LEADERSHIP

Chair , Graduate Assistant Advisory Committee (GAAC), UMBC Graduate Student Association	2026 – 2027
Lead Volunteer , ACM/IEEE Symposium on Edge Computing (SEC 2025)	Dec. 2025
HR Head , ISTE Students' Chapter, Manipal Institute of Technology	2017 – 2018
Co-Event Head (Acumen), TechTatva '17, Manipal Institute of Technology	Oct. 2017
Management Committee Member , ISTE Students' Chapter, Manipal Institute of Technology	2016 – 2017

TECHNICAL SKILLS

Languages	Swift, Python, Java, Shell, SQL, HTML/CSS
Methods	Acoustic and ultrasonic signal processing, physiological sensing, tissue biomechanics, embedded audio, machine learning
Tools	Xcode, SwiftUI, Android Studio, Git, GitLab CI/CD, L ^A T _E X

REFERENCES

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